

ASSIGNMENT-1.5

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BATCH: 30

Task 1 :

PROMPT:

Write a Python program that takes a string as input and reverses it without using any user-defined functions.

CODE:

```
1 #generate string reversal using without function
2 def reverse_string(s):
3     reversed_str = ""
4     for char in s:
5         reversed_str = char + reversed_str
6     return reversed_str
7 input_string = "hello"
8 print("Original string:", input_string)
9 print("Reversed string:", reverse_string(input_string))
```

OUTPUT:

```
sharanyareddy@Sharanyas-MacBook-Air ~ % /usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task1.py"
Original string: hello
Reversed string: olleh
sharanyareddy@Sharanyas-MacBook-Air ~ %
```

OBSERVATION:

GitHub Copilot created a simple program to reverse a string. The code works correctly and takes user input. The logic is easy to understand but is written in one place without using functions.

Task 2 :

PROMPT:

Simplify this string reversal code and improve readability.

CODE:

```
Task2.py > ...
1  # String reversal without using any function
2  input_string = "hello"
3  reversed_str = ""
4  for char in input_string:
5      reversed_str = char + reversed_str
6  print("Original string:", input_string)
7  print("Reversed string:", reversed_str)
8  #output: !dlrow ,olleh
```

OUTPUT:

```
/usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task2.py"
sharanyareddy@Sharanyas-MacBook-Air AI-Assist coding % /usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task2.py"
Original string: hello
Reversed string: olleh
sharanyareddy@Sharanyas-MacBook-Air AI-Assist coding %
```

OBSERVATION:

Copilot improved the code by making it shorter and cleaner. Unnecessary variables were removed and slicing was used to reverse the string. This made the program faster and easier to read.

Task 3:

PROMPT:

Write a Python function that takes a string and returns the reversed string with comments.

CODE:

```

Task3.py > ...
1  #generate a string reversal using function
2  num = 5
3  factorial = 1
4  for i in range(1, num + 1):
5      factorial *= i
6  print("The factorial of", num, "is:", factorial)

```

OUTPUT:

```

/usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task3.py"
sharanyareddy@Sharanyas-MacBook-Air ~ % /usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task3.py"
The factorial of 5 is: 120
sharanyareddy@Sharanyas-MacBook-Air ~ %

```

OBSERVATION:

Copilot generated a function to reverse a string. Using a function makes the code more organized and reusable. The program becomes easier to understand and maintain.

Task 4:

Aspect	Without Functions	With Functions
Code Clarity	Code is written in one place and can look messy	Code is neat and easy to understand
Reusability	Same code must be written again	Same function can be used many times
Debugging	Hard to find and fix errors	Easy to fix errors in one place
Use in Big Programs	Not good for large programs	Very good for large programs
Maintenance	Takes more time to update	Easy to update and manage

Task 5:

PROMPT:

Generate two string reversal implementations: loop-based and slicing-based.

CODE:

```
Task5.py > ...
1  #AI-Generated Iterative vs Recursive Fibonacci Approaches (Different Algorithmic Approaches to String Reversal)
2  def fibonacci_reverse(s):
3      n = len(s)
4      fib = [0, 1]
5      for i in range(2, n + 1):
6          fib.append(fib[-1] + fib[-2])
7
8      reversed_str = ""
9      for i in range(n - 1, -1, -1):
10         reversed_str += s[i]
11     return reversed_str
12
13 test_string = "hello"
14 print("Fibonacci approach - Original:", test_string)
15 print("Fibonacci approach - Reversed:", fibonacci_reverse(test_string))
```

OUTPUT:

```
/usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task5.py"
sharanyareddy@Sharanyas-MacBook-Air ~ % /usr/local/bin/python3 "/Users/sharanyareddy/AI-Assist coding/Task5.py"
Fibonacci approach - Original: hello
Fibonacci approach - Reversed: olleh
sharanyareddy@Sharanyas-MacBook-Air ~ %
```

OBSERVATION:

Copilot provided two ways to reverse a string. The loop method shows the step-by-step process but is slower. The slicing method is faster and simpler, so it is better for real applications.